

Problem Set 5: Metric Spaces

Due Friday

Exercise 1. Let $d(x, y) = |x - y|$ on $\mathbb{R}$. Show that d is a metric.

Solution. Positivity and symmetry are clear. For the triangle inequality,

$$d(x, z) = |(x - y) + (y - z)| \leq |x - y| + |y - z| = d(x, y) + d(y, z).$$

□

Exercise 2. Let $\|\cdot\|$ be a norm on a vector space V and define $D(u, v) := \|u - v\|$.

(a) Prove that D is a metric on V .

(b) Prove that $|\|u\| - \|v\|| \leq \|u - v\|$.

Solution.

(a) The norm axioms give $D(u, v) = 0 \iff u = v$ and $D(u, v) = \|-(v - u)\| = D(v, u)$; the triangle inequality for the norm gives the triangle inequality for D .

(b) Write $u = (u - v) + v$ to get $\|u\| \leq \|u - v\| + \|v\|$, and swap the roles of u and v .

□

Exercise 3. Show that the discrete metric

$$\delta(x, y) = \begin{cases} 0 & \text{if } x = y, \\ 1 & \text{otherwise} \end{cases}$$

makes every subset of a set X open.

Solution. For $x \in U$ the open ball $B(x, \frac{1}{2}) = \{x\}$ lies in U .

□

Exercise 4. Prove that a sequence in a metric space has at most one limit.

Solution. If $x_n \rightarrow a$ and $x_n \rightarrow b$, then for every $\varepsilon > 0$ there is N with $d(a, b) \leq d(a, x_N) + d(x_N, b) < 2\varepsilon$. Hence $d(a, b) = 0$ and $a = b$.

□